Optimisation

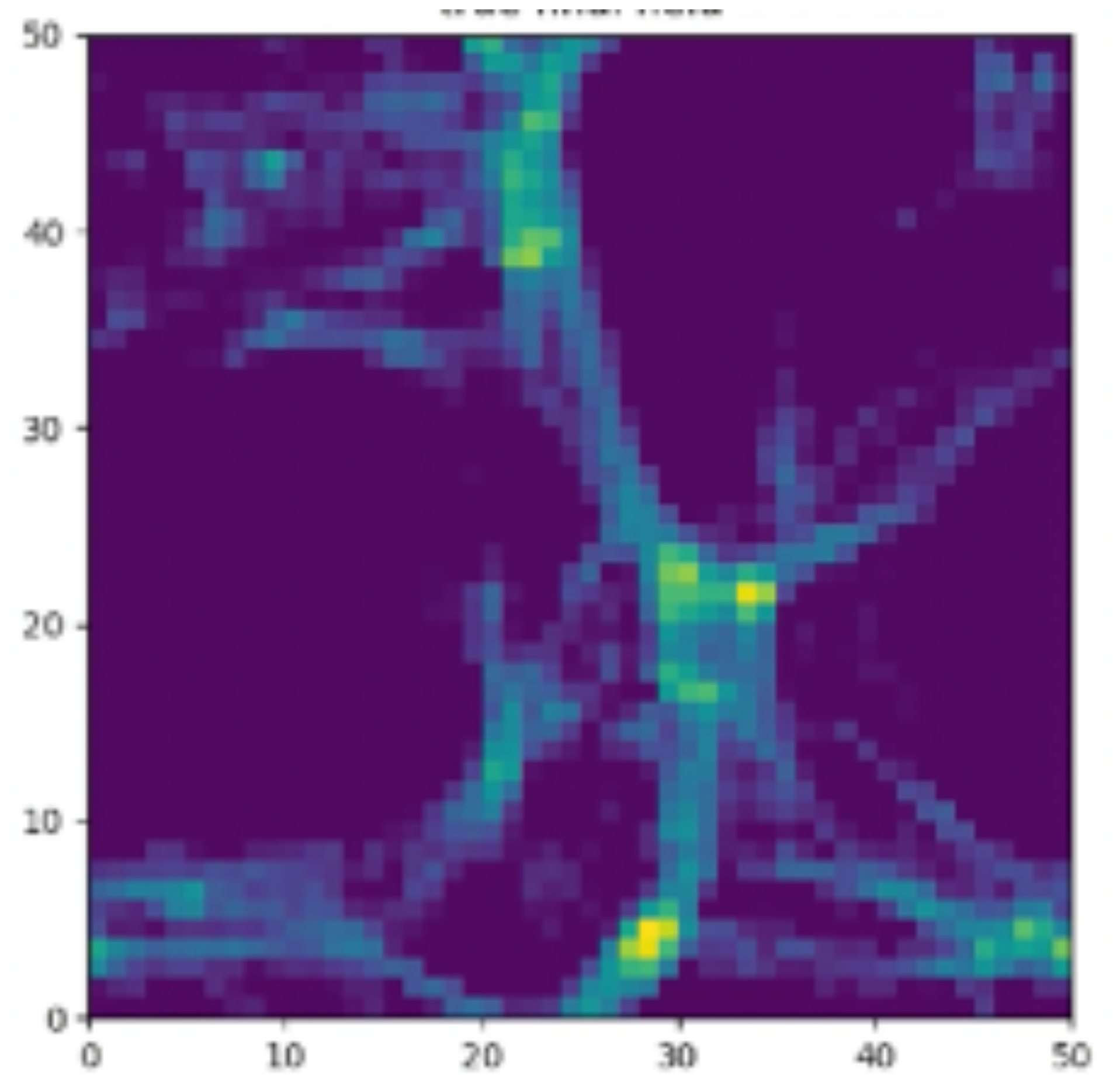
9

Observations : Galaxy catalogues → galaxy 3D density maps

Aim : Infer current and past *dark matter* density map from current *galaxy* density map

Method : Using `JAX / Tensorflow`, optimise differentiable N-body simulations (gravitation only ≡ dark matter)

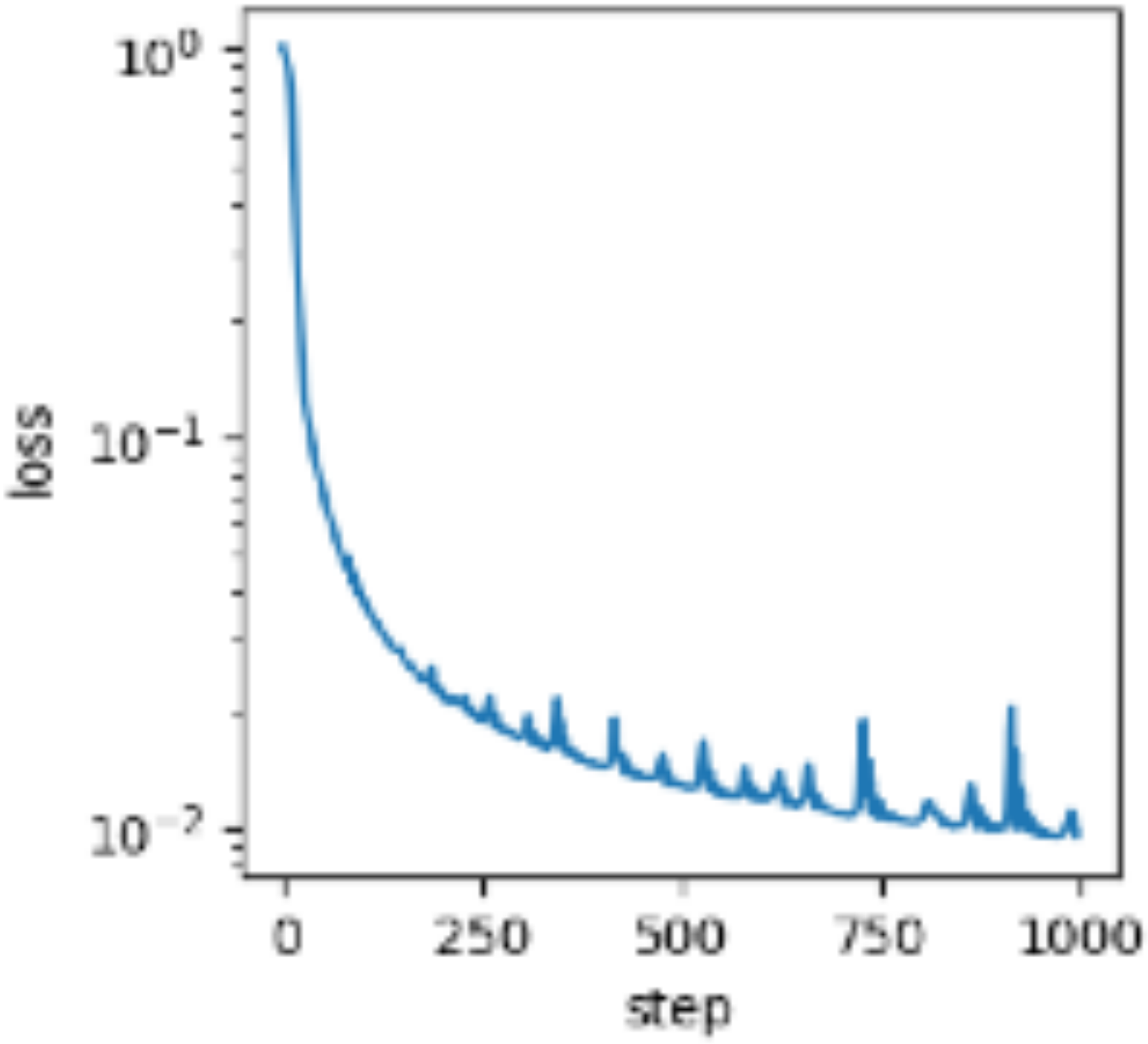
The combined speed and differentiability allows to run optimisation and reconstruct the structure formation history from late-time observations (local Universe)


50
40
30
20
10
0
0
10
20
30
40
50

Observation

Optimisation (ADAM)

Here I show a simple test implementation of such an optimisation.

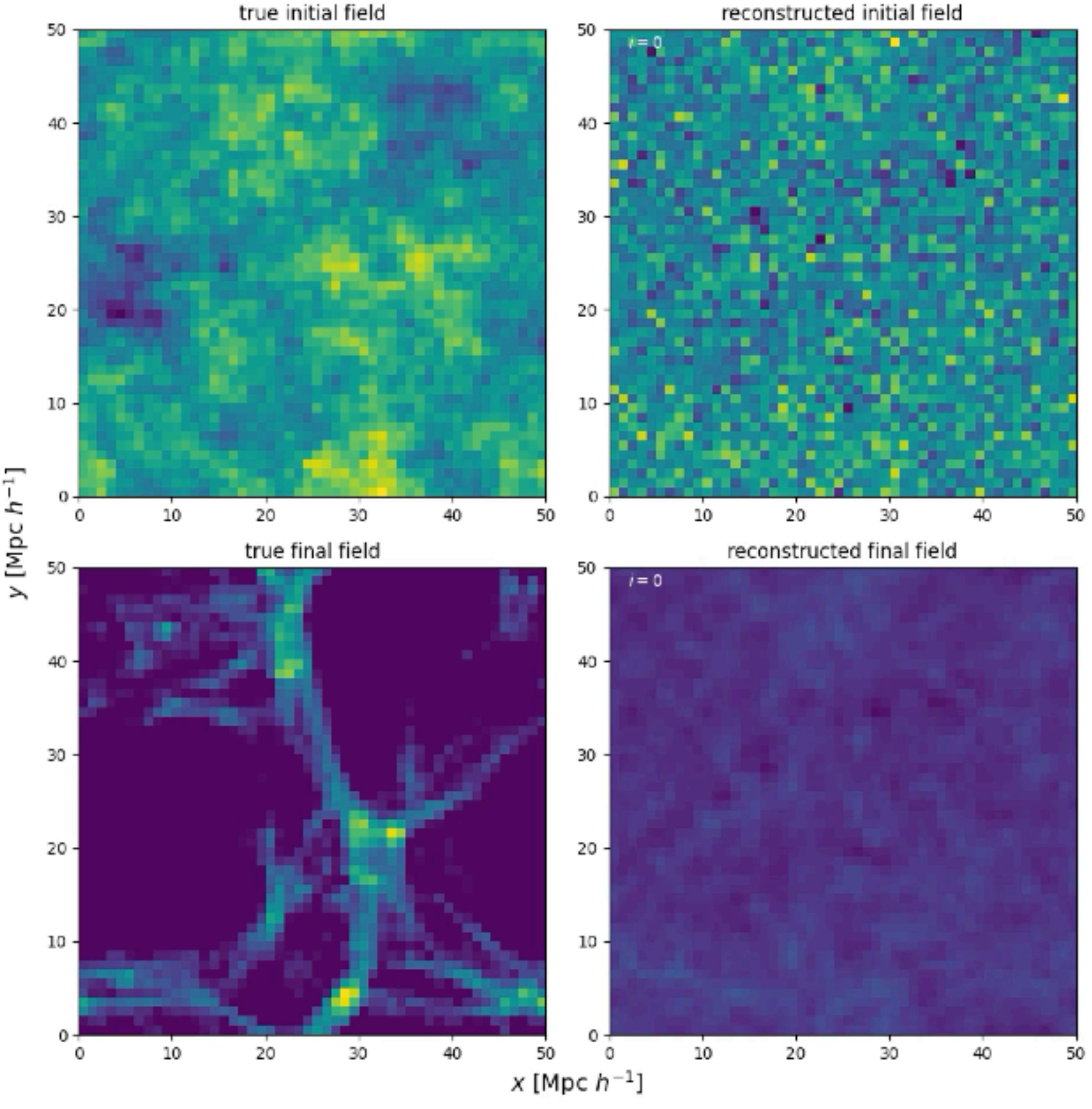

10^0
10^{-1}
10^{-2}
loss
0
250
500
750
1000
step

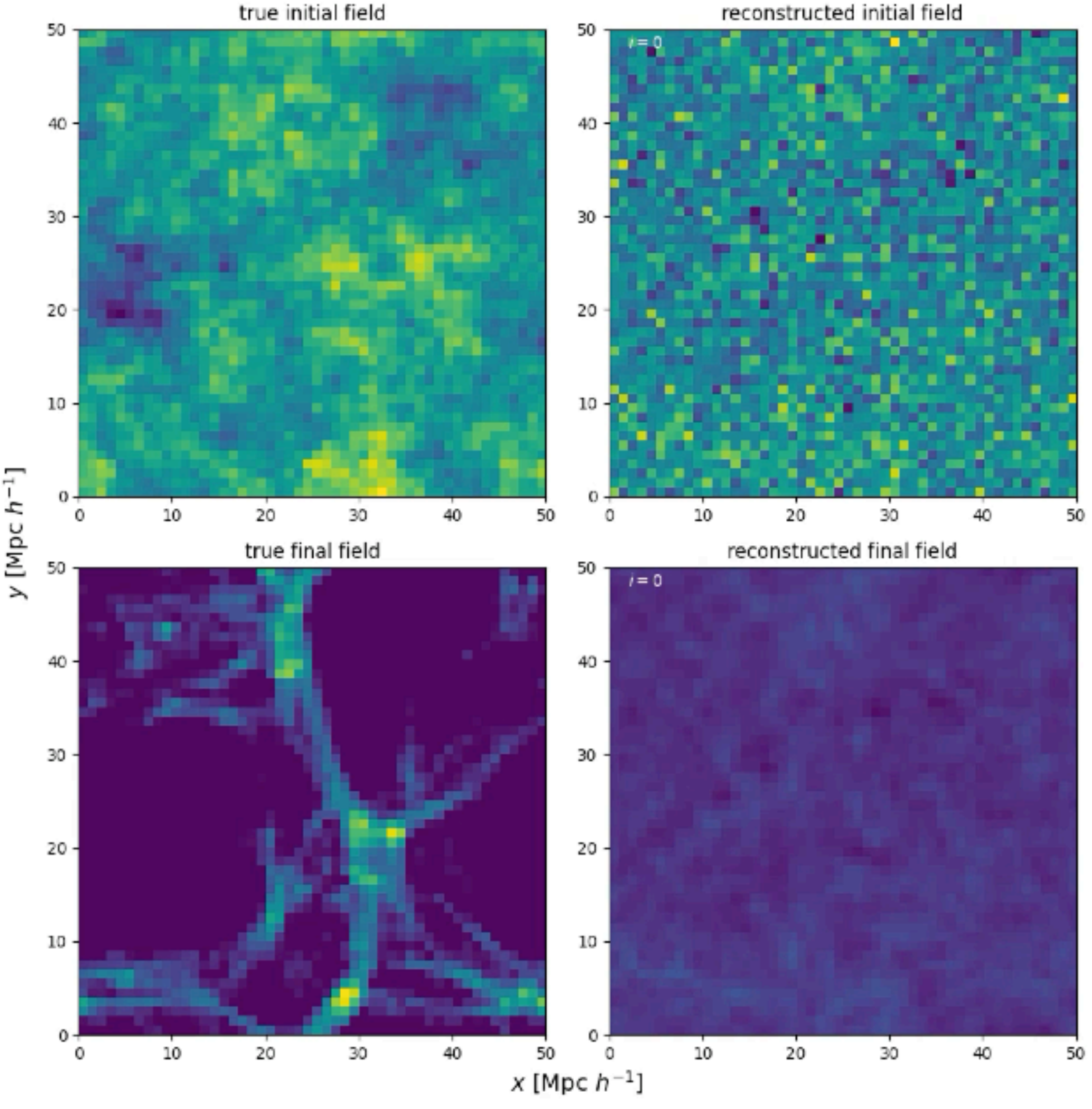
Loss

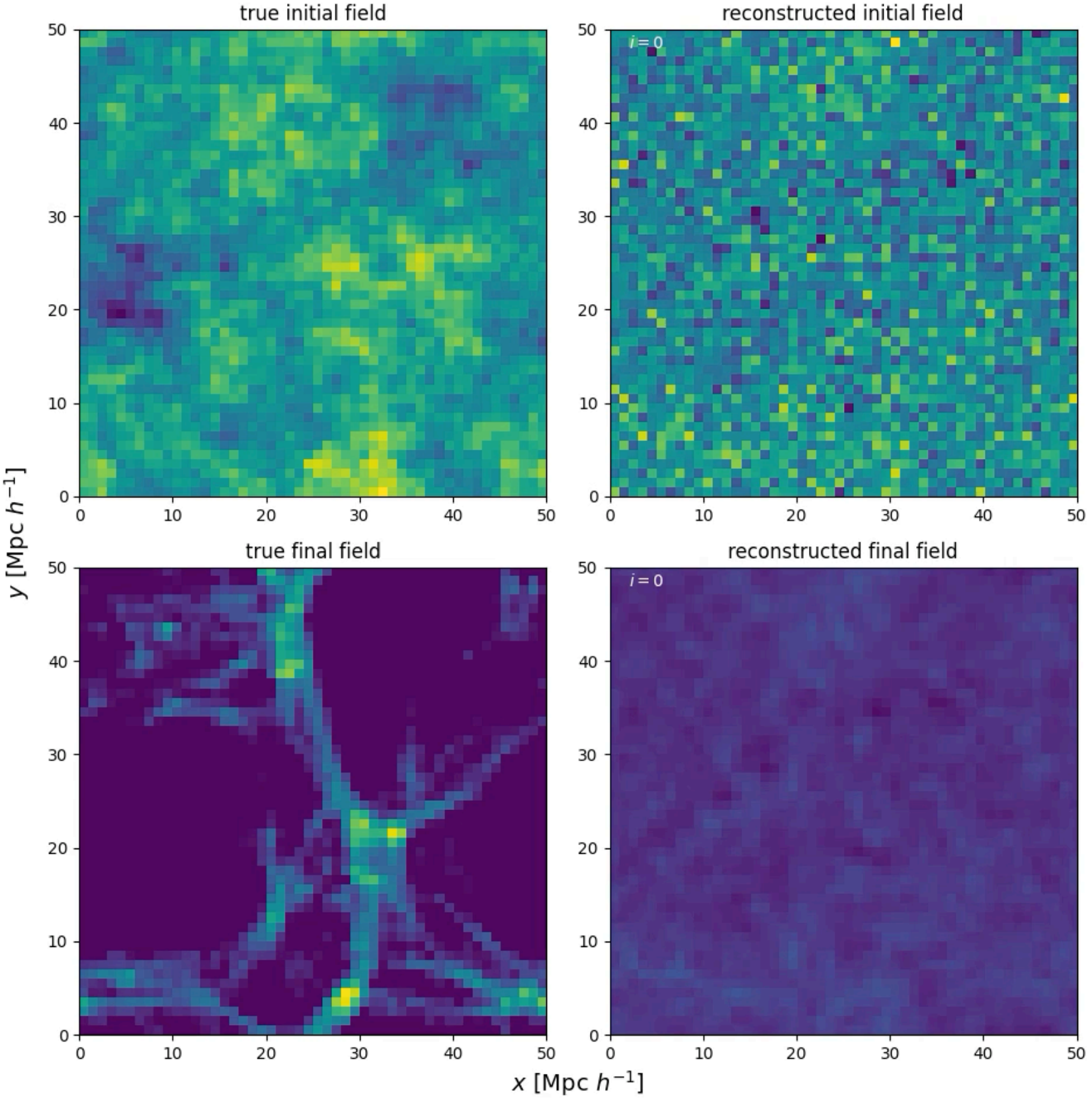
Initial conditions (early-time)

Final density field (late-time)

We see the constraints on the density field getting better through the optimisation


true initial field
reconstructed initial field
true final field
reconstructed final field
$i = 0$
$i = 0$
y [Mpc h^{-1}]
x [Mpc h^{-1}]
0
10
20
30
40
50


true initial field
reconstructed initial field
true final field
reconstructed final field
$i = 0$
$i = 0$
y [Mpc h^{-1}]
x [Mpc h^{-1}]
0
10
20
30
40
50


true initial field
reconstructed initial field
true final field
reconstructed final field
$i = 0$
$i = 0$
y [Mpc h^{-1}]
x [Mpc h^{-1}]
0
10
20
30
40
50

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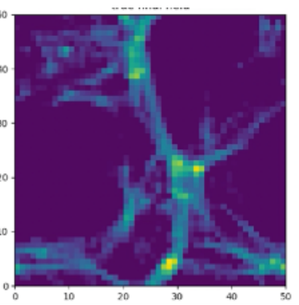
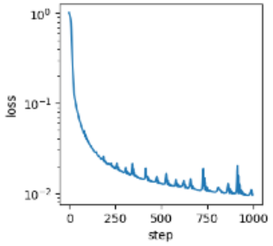
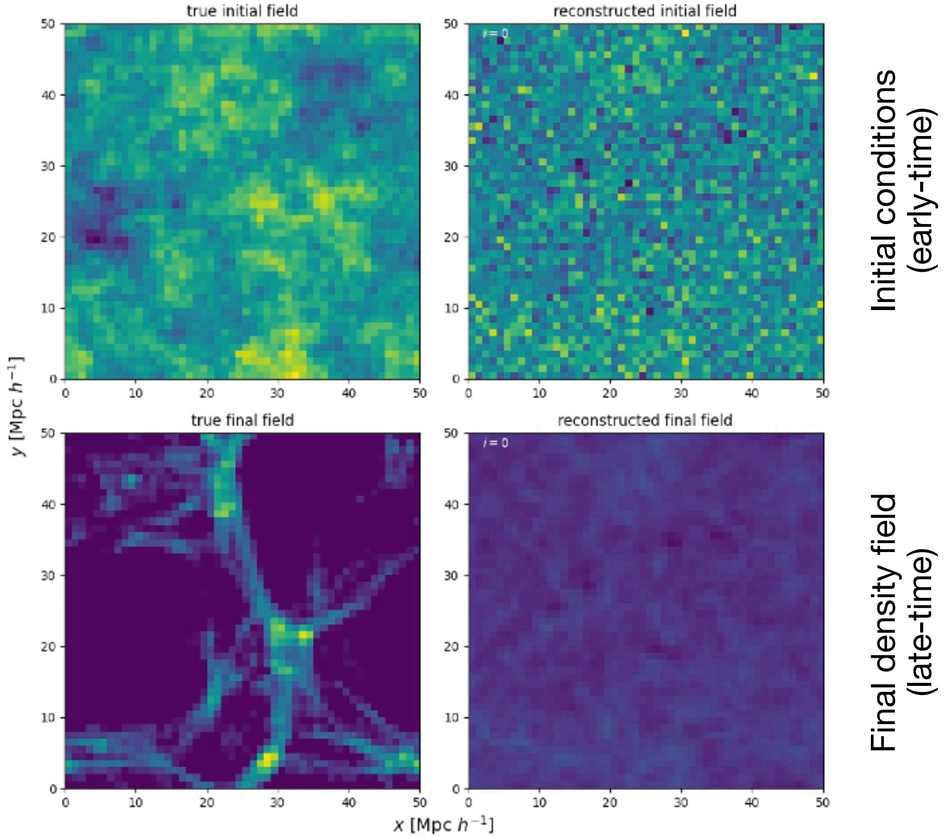


Loss



Optimisation (ADAM)



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